

SIMATS
ONLINE
LAB
ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMINGYUGITHA B
192424102RD**QUESTIONS****10 / 10 completed**

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted

by row then col), and display dense matrix form. Max 10x10; ignore zeros.

Sample Input (Matrix 1):

3 3 3

0 1 5

1 0 8

2 2 2

Sample Input (Matrix 2):

3 3 2

0 1 3

2 2 7

Sample Output (Sum Dense):

0 8 0

8 0 0

0 0 9

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 #define MAX 10
4 #define MAX_TERMS 100
5 typedef struct {
6     int r, c, v;
7 } Triple;
8
9 static void sortByRC(Triple a[], int n) {
10
11     for (int i = 0; i < n - 1; i++) {
12         for (int j = i + 1; j < n; j++) {
13             if (a[j].r < a[i].r || (a[j].r == a[i].r && a[j].c <
14                 Triple t = a[i];
15                 a[i] = a[j];
16                 a[j] = t;
17             }
18         }
19     }
20 }
21
```

OUTPUT

```
0 5 0
8 0 0
0 0 2
```

INPUT

3 3 3

0 1 5

1 0 8

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